

Recent BS Chemical Engineering Course Revisions:

EPC Meeting approval in March 2025

1. CHE 101: Introduction to Chemical Engineering & Sustainability

Scope of Revisions: Title change, updated topics

This course provides an overview of the principles and unit operations of chemical engineering as practiced in traditional industries and in sustainable technologies. Five key sustainable themes are introduced: sustainable energy, sustainable health, sustainable food, sustainable air, and sustainable water. Students will explore the challenges in these areas and the role of chemical engineering in developing sustainable products and processes. The topics covered will provide context to subsequent 200 and 200 level chemical engineering courses and the connection of chemical engineering to basic sciences and other engineering disciplines.

In progress: Academic proposal to add CHE 101 as a required course in the BS ChE degree

2. CHE 426: Biopharmaceutical Process Development & Manufacturing

Scope of Revisions: Make a permanent 400-level technical elective course

A course entitled “Biopharmaceutical Process Development and Manufacturing” has been taught every Spring semester since 2021 under the general ChE 494 (Selected Topics in Chemical Engineering) rubric. The course provides an in-depth view into the various aspects involved in process development and process manufacturing in the pharmaceutical and biotechnology industry. The course has been very well received by our students due to increasing interest in biotechnology. Thus, the ChE Department wishes to offer the course under its own rubric, number and name.

3. CHE 427 Entrepreneurship in Engineering and CHE 451 Renewable Energy

Scope of Revisions:

Revising the Type of Course description to be equivalent to other the 400-level courses that are both a requirement for a concentration and a technical elective for the BS degree

4. CHE 499: Professional Development Seminar

Scope of Revisions: Revising course description and topics and updating when course offered (Fall semester only)